Photometric Test Report

Reference Standards

UL1598-2008

ANSI C82.77-10-2014

IES LM-79-2019

Prepared For

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Test Laboratory:

UL-CCIC Company Limited

Test Laboratory Address:

No.2, Chengwan Road, Suzhou Industrial Park, Suzhou 215122, China

Catalog Number

BPANEL-75S5-UNV-D-8SC61-24-G-WH-H

Project Number

4791498147.3

Report Number

4791498147.3-10a

Test Date

2024-10-30~2024-11-07

Issue Date

2024-11-21

Revision Date

N/A

Prepared By	**Approved By**
Becky Fan	Roger Xu
Becky Fan	Roger Xu

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Test List

Sample Received Date: 2024-10-16

Test Item	Sample ID	Model Number	Tests Conducted By
Integrating Sphere Test	7712181	BPANEL-75S5-UNV-D-8SC61-24-G-WH-H	Hom Dong
Goniophotometer Test	7712181	BPANEL-75S5-UNV-D-8SC61-24-G-WH-H	Hom Dong
THD and PF Test	7712181	BPANEL-75S5-UNV-D-8SC61-24-G-WH-H	Hom Dong
In-Situ Temperature Measurement Test	7712181	BPANEL-75S5-UNV-D-8SC61-24-G-WH-H	Hom Dong

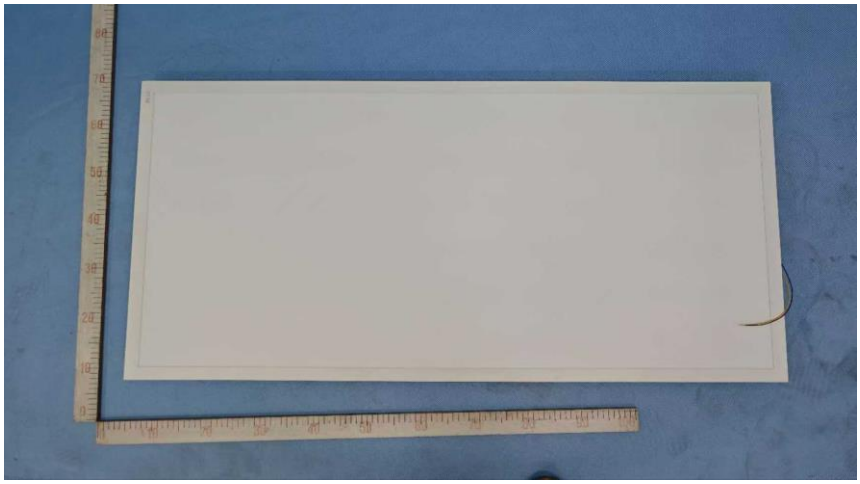
Remark (if any)

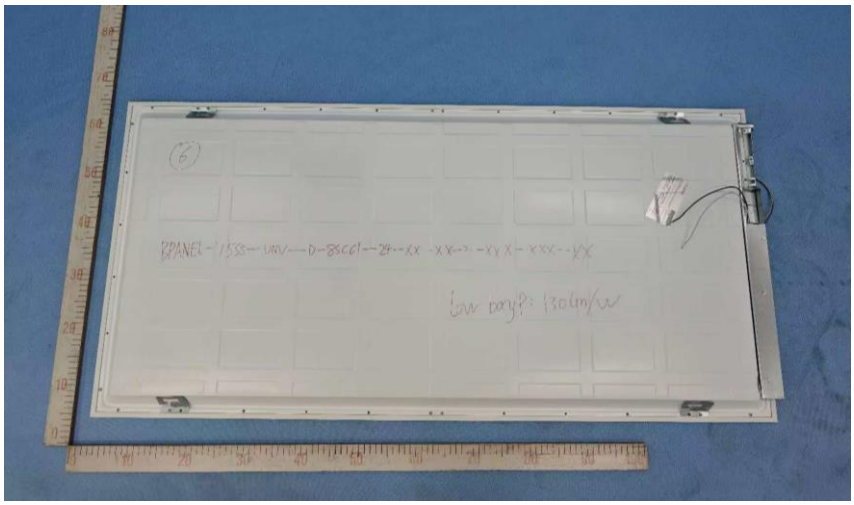
1. UL test equipment information is recorded on Meter Use in UL's Aurora database.

Product Description

Luminaire Description: Low Bay Luminaires (Commercial and Industrial)
Model Number: BPANEL-75S5-UNV-D-8SC61-24-G-WH-H
Electrical Ratings and CCT: 120-277Vac, 50/60 Hz, 75W max, 3000K/3500K/4000K/5000K/5700K/6500K tunable
Driver Model Number: BQ-65L6-1470-44-M56
LED Package: BXVN-30E-13H-9NV, BXVN-65E-13H-9NV, Bridgelux Inc
Family Model and Variation: BPANEL-75S5-UNV-D-8SC61-24-XX-YY-H-ZZZ-***-##, where "XX" represent mounting, can be G=Grid mounting, SF=Surface mounting, SP=Suspended mounting, DF=Drywall flange mounting. "YY" represent Finish color, can be WH=White or others. "ZZZ" represent Emergency, can be EM*= Emergency types(* can be blank or any letter), blank=No emergency. "***" represent Sensor receptacle, can be SR#= Sensor receptacle types(# can be blank or any number), blank= Without sensor receptacle. ## represent junction box, can be JB=with junction box, blank=without junction box.

Photos of Products Characteristics





Integrating Sphere Test

Model No.	BPANEL-75S5-UNV-D-8SC61-24-G-WH-H		Sample ID.	7712181
Operate time (Min.)	55	Stabilization time (Min.)		50

Test Method

1.The sample was tested according to the IES LM-79-2019, and the product is assume to be brand new without seasoning.
2.Photometric paramters were measured using an integrating sphere, a spectroradiometer and software. The ambient temperature condition inside the sphere was maintained at 25 °C ± 0.9 °C.The reference standard lamp is power 100W omni-directional Incandescent lamp and was calibrated by National Institute of Metrology, China.
3.The sample measurements were made using a spectroradiometer connected by a fiber optic cable and detector through the detector port of the integrating sphere. Coating reflectance of the integrating sphere was 90% to 98%. Photometric measurement conditions was using 4π geometry. The self-absorption factor is applied in the final test result. The sample was operated at rated voltage and was stabilized before measurement. Chromaticity coordinates, correlated color temperature and color rendering index were calculated from the spectral radiant flux measurements taken at 1 nm intervals over the range of 380 to 780 nm.

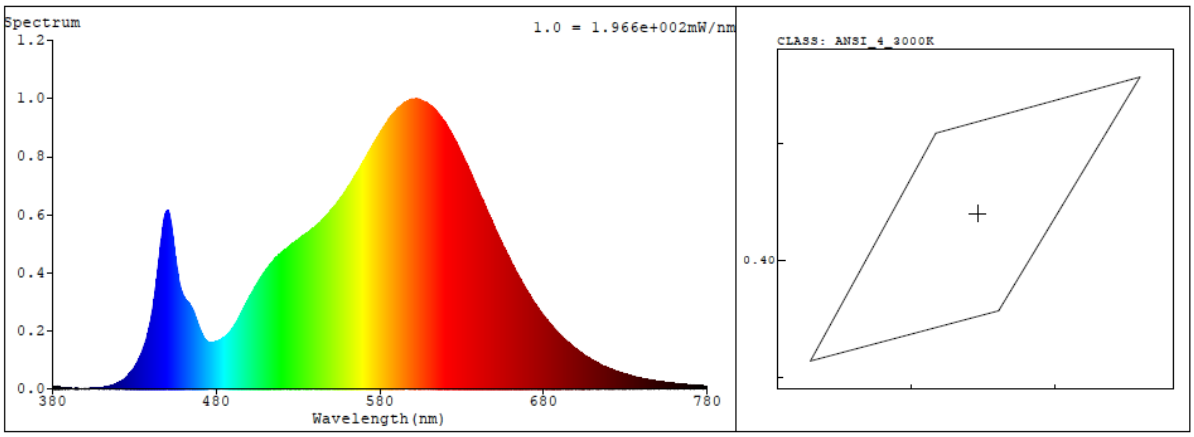
Integrating Sphere Test Conditions

Temperature (°C)	Voltage (Vac)	Frequency (Hz)	Current (A)	Power (W)	Power Factor	Orientation
25.1	120	60	0.589	70.17	0.9932	Horizontal

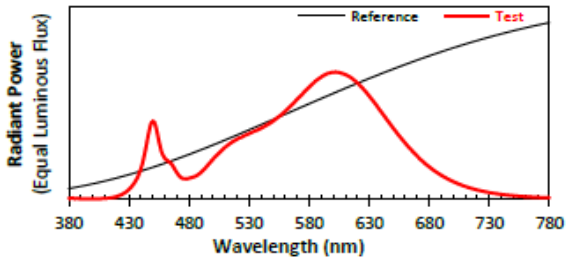
Test Results

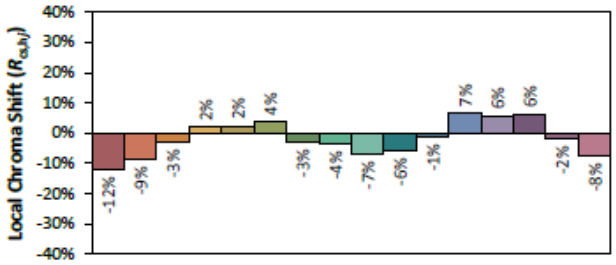
CCT (K)	CRI (Ra)	R9	Rf	Rg	Flux (lm)	Luminous Efficacy (lm/W)
3037	81	2	84	96	9386.6	133.8

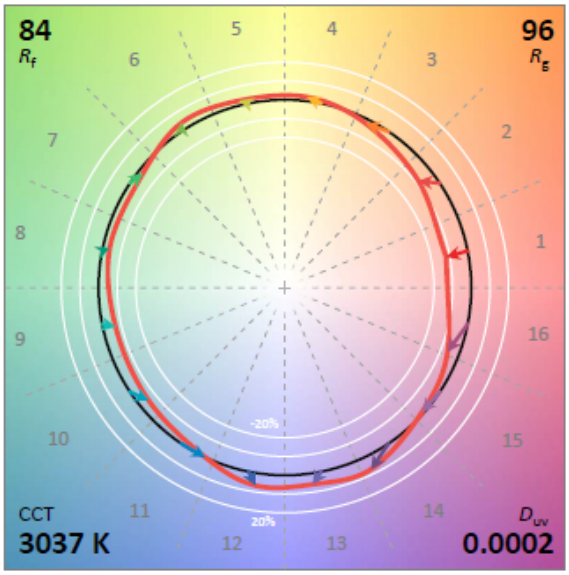
Spectrum

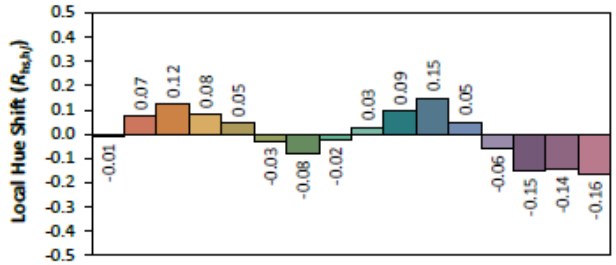


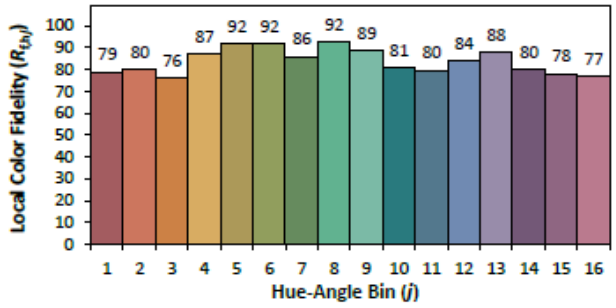
ANSI/IES TM-30-18 Color Rendition Report

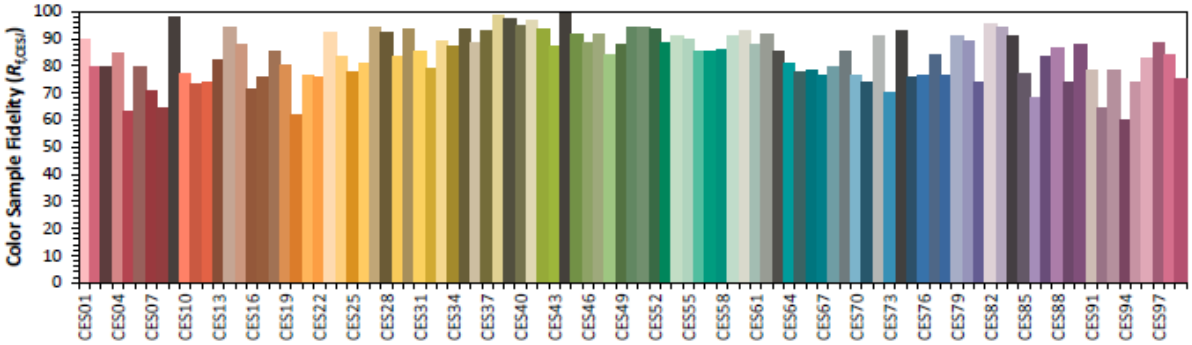












Notes: This is a recommended method for displaying ANSI/IES TM-30-18 information.

x	0.4347
y	0.4039
u'	0.2492
v'	0.5210

CIE 13.3-1995 (CRI)	
R_a	81
R_9	2

Colors are for visual orientation purposes only. Created with the ANSI/IES TM-30-18 Calculator Version 2.00.

Integrating Sphere Test

Model No.	BPANEL-75S5-UNV-D-8SC61-24-G-WH-H	Sample ID.	7712181
Operate time (Min.)	55	Stabilization time (Min.)	50

Test Method

1.The sample was tested according to the IES LM-79-2019, and the product is assume to be brand new without seasoning.
2.Photometric paramters were measured using an integrating sphere, a spectroradiometer and software. The ambient temperature condition inside the sphere was maintained at 25 °C ± 0.9 °C.The reference standard lamp is power 100W omni-directional Incandescent lamp and was calibrated by National Institute of Metrology, China.
3.The sample measurements were made using a spectroradiometer connected by a fiber optic cable and detector through the detector port of the integrating sphere. Coating reflectance of the integrating sphere was 90% to 98%. Photometric measurement conditions was using 4π geometry. The self-absorption factor is applied in the final test result. The sample was operated at rated voltage and was stabilized before measurement. Chromaticity coordinates, correlated color temperature and color rendering index were calculated from the spectral radiant flux measurements taken at 1 nm intervals over the range of 380 to 780 nm.

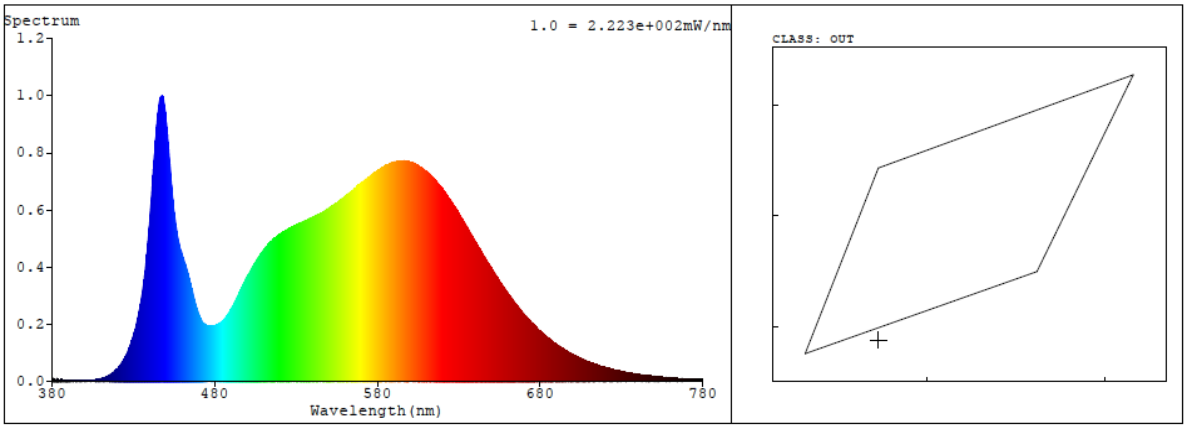
Integrating Sphere Test Conditions

Temperature (°C)	Voltage (Vac)	Frequency (Hz)	Current (A)	Power (W)	Power Factor	Orientation
25.1	120	60	0.585	69.68	0.9934	Horizontal

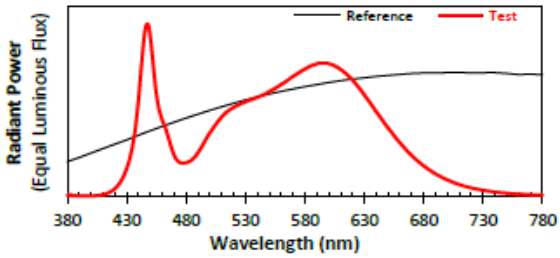
Test Results

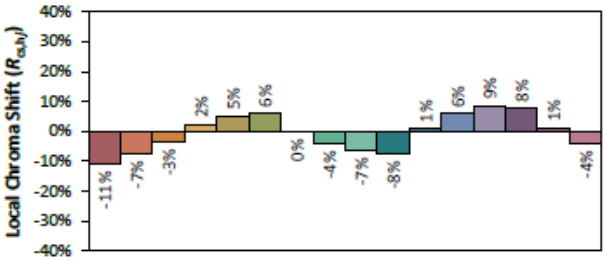
CCT (K)	CRI (Ra)	R9	Rf	Rg	Flux (lm)	Luminous Efficacy (lm/W)
4029	84	13	84	98	9539.3	136.9

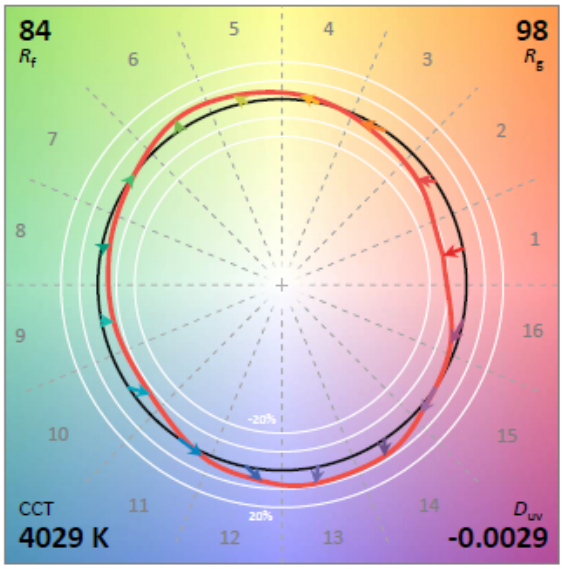
Spectrum

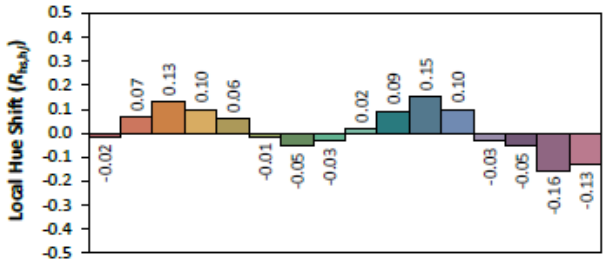


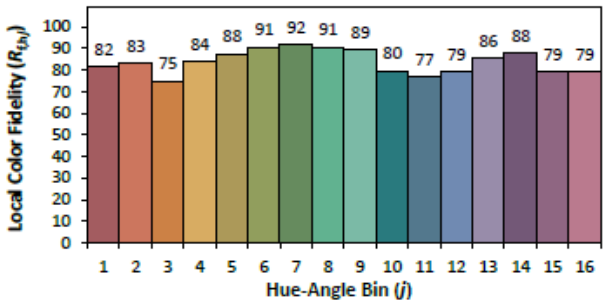
ANSI/IES TM-30-18 Color Rendition Report

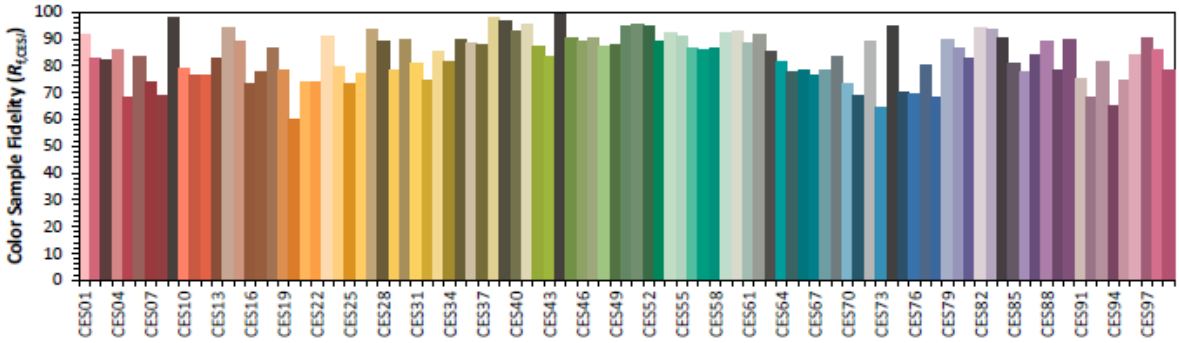












Notes: This is a recommended method for displaying ANSI/IES TM-30-18 information.

x	0.3772
y	0.3686
u'	0.2263
v'	0.4975

CIE 13.3-1995 (CRI)	
R_a	84
R_9	13

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Integrating Sphere Test

Model No.	BPANEL-75S5-UNV-D-8SC61-24-G-WH-H	Sample ID.	7712181
Operate time (Min.)	55	Stabilization time (Min.)	50

Test Method

1.The sample was tested according to the IES LM-79-2019, and the product is assume to be brand new without seasoning.
2.Photometric paramters were measured using an integrating sphere, a spectroradiometer and software. The ambient temperature condition inside the sphere was maintained at 25 °C ± 0.9 °C.The reference standard lamp is power 100W omni-directional Incandescent lamp and was calibrated by National Institute of Metrology, China.
3.The sample measurements were made using a spectroradiometer connected by a fiber optic cable and detector through the detector port of the integrating sphere. Coating reflectance of the integrating sphere was 90% to 98%. Photometric measurement conditions was using 4π geometry. The self-absorption factor is applied in the final test result. The sample was operated at rated voltage and was stabilized before measurement. Chromaticity coordinates, correlated color temperature and color rendering index were calculated from the spectral radiant flux measurements taken at 1 nm intervals over the range of 380 to 780 nm.

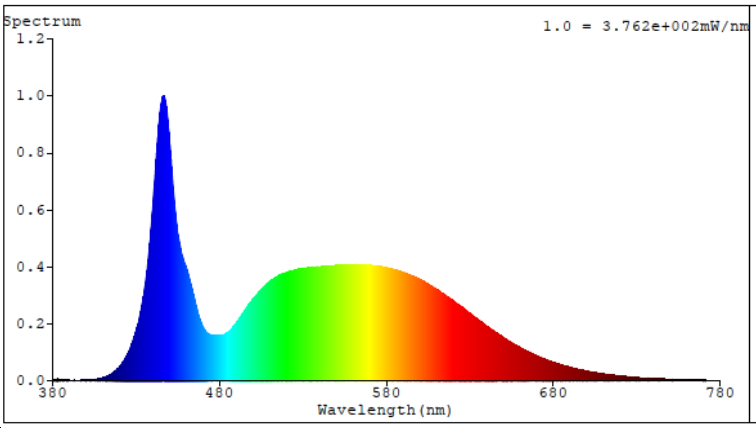
Integrating Sphere Test Conditions

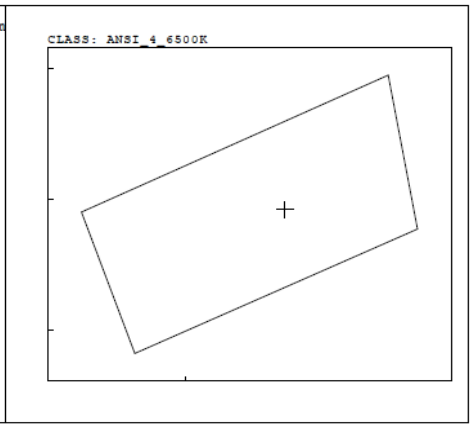
Temperature (°C)	Voltage (Vac)	Frequency (Hz)	Current (A)	Power (W)	Power Factor	Orientation
25.1	120	60	0.586	69.87	0.9933	Horizontal

Test Results

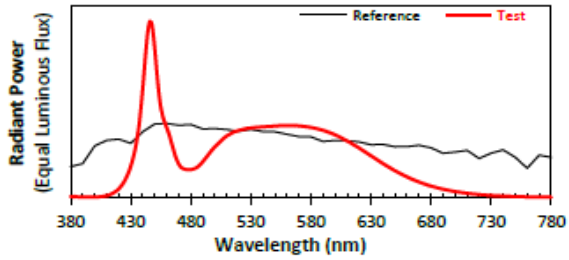
CCT (K)	CRI (Ra)	R9	Rf	Rg	Flux (lm)	Luminous Efficacy (lm/W)
6453	81	1	81	97	9560.8	136.8

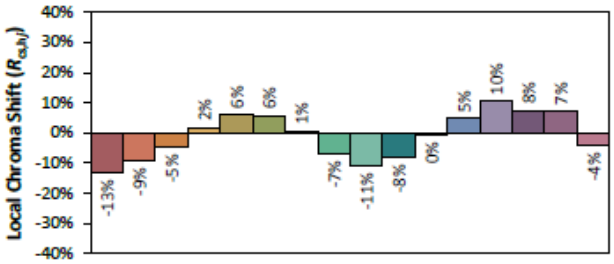
Spectrum

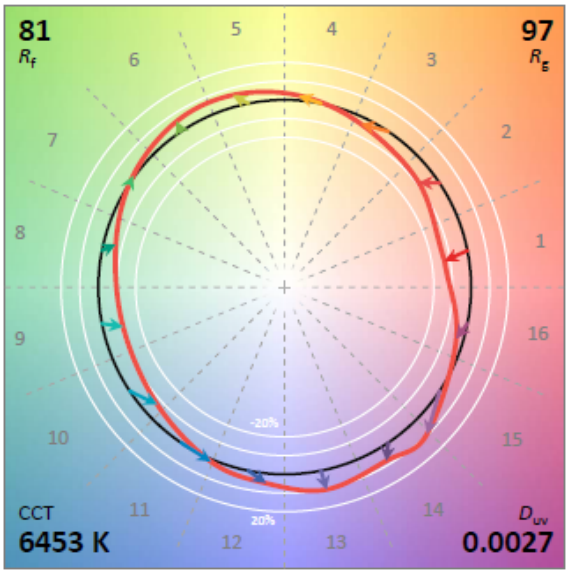


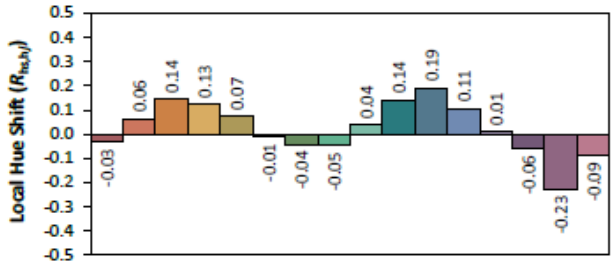


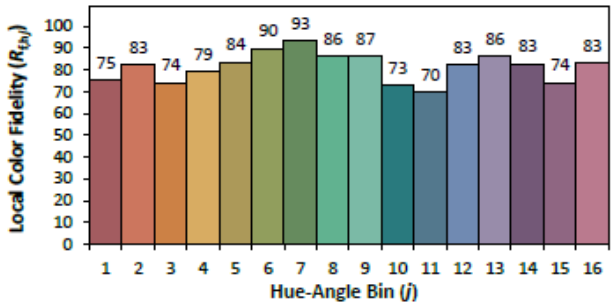
ANSI/IES TM-30-18 Color Rendition Report

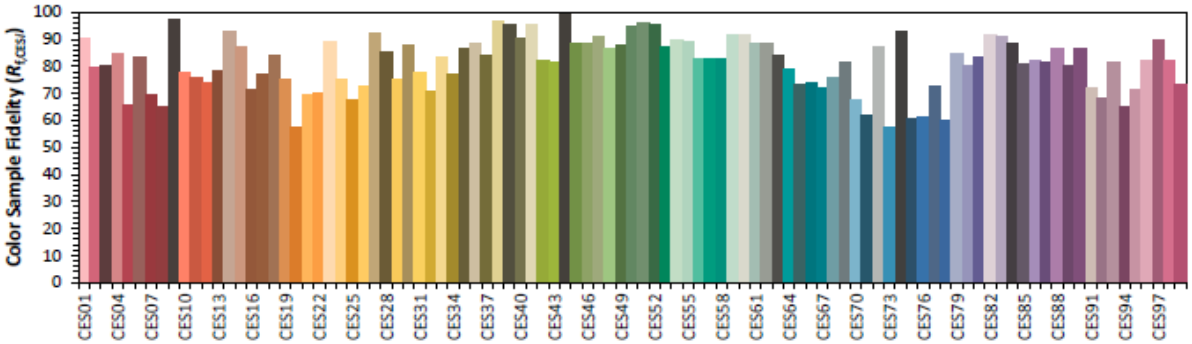












Notes: This is a recommended method for displaying ANSI/IES TM-30-18 information.

x	0.3137
y	0.3290
u'	0.1985
v'	0.4685

CIE 13.3-1995 (CRI)	
R_a	81
R_9	1

Colors are for visual orientation purposes only. Created with the ANSI/IES TM-30-18 Calculator Version 2.00.

Goniophotometer Test

Model No.	BPANEL-75S5-UNV-D-8SC61-24-G-WH-H	Sample ID.	7712181
Operate time (Min.)	65	Stabilization time (Min.)	50

Test Method

1.The sample was tested according to the IES LM-79-2019, and the product is assume to be brand new without seasoning.
2.Photometric parameters were measured using a type C goniophotometer and software.
3.The ambient temperature shall be maintained at 25 °C ± 0.9 °C, measured at a point not more than 1.5 m from the sample and at the same height as the sample. The reference standard lamp is power 400W omni-directional Incandescent lamp and was calibrated by National Institute of Metrology, China.
4.The samples were operated at rated voltage and was stabilized before measurement. Luminous flux, luminaire efficacy, zonallumen were calculated from the software taken at 1° vertical intervals and 22.5° horizontal intervals. Photometric distance was more than five times of the largest dimension of the test SSL product.

Goniophotometer Test Conditions

Temperature (°C)	Voltage (Vac)	Frequency (Hz)	Current (A)	Power (W)	Power Factor	Orientation
24.5	120.09	60	0.588	70.26	0.995	Horizontal

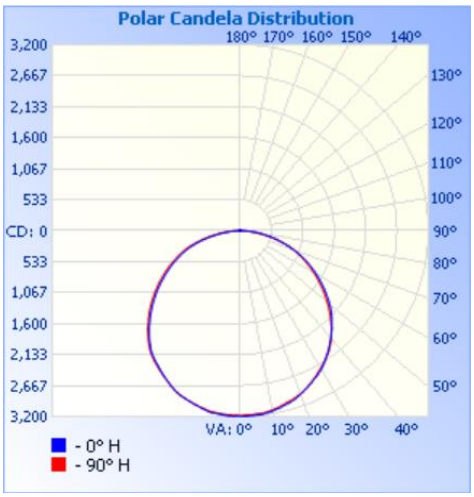
Test Results

Zonal Lumen Requirement (20°-50°)	Field Angle (10%)		Beam Angle (50%)		Flux (lm)	Luminous Efficacy (lm/W)
	Horizontal Spread	Vertical Spread	Horizontal Spread	Vertical Spread		
48.5%	165.3	165.5	114.9	114.9	9382.7	133.5

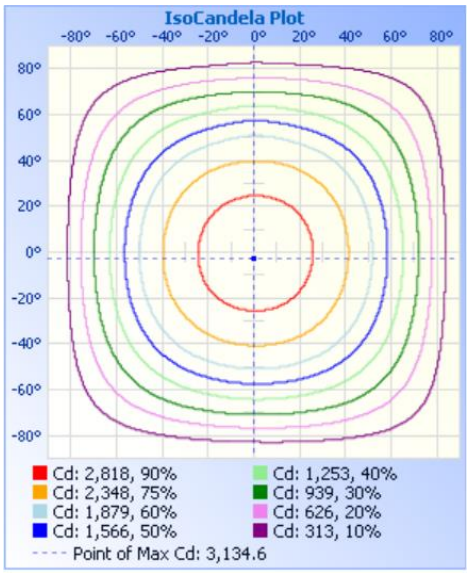
UGR
22.2

Goniophotometer Test (Cont'd)

Polar Candela Distribution



IsoCandela Plot



Goniophotometer Test (Cont'd)

Zonal Lumen Summary

Zonal Lumen Summary		
Zone	Lumens	% Luminaire
0-30	2,450.5	26.1%
0-40	4,043.5	43.1%
0-60	7,195.5	76.7%
60-90	2,165.0	23.1%
70-100	982.1	10.5%
90-120	10.9	0.1%
0-90	9,360.4	99.8%
90-180	22.3	0.2%
0-180	9,382.7	100%

Lumens Per Zone

Lumens Per Zone					
Zone	Lumens	% Total	Zone	Lumens	% Total
0-10	296.3	3.2%	90-100	5.5	0.1%
10-20	852.0	9.1%	100-110	2.7	0%
20-30	1,302.2	13.9%	110-120	2.7	0%
30-40	1,593.0	17.0%	120-130	2.6	0%
40-50	1,655.2	17.6%	130-140	2.6	0%
50-60	1,496.8	16.0%	140-150	2.4	0%
60-70	1,188.3	12.7%	150-160	1.9	0%
70-80	747.8	8.0%	160-170	1.4	0%
80-90	228.8	2.4%	170-180	0.6	0%

Goniophotometer Test (Cont'd)

Intensity Data(cd)

	0	22.5	45	67.5	90	112.5	135	157.5	180	202.5	225	247.5	270	292.5	315	337.5	360
0	3125	3125	3125	3125	3125	3125	3125	3125	3125	3125	3125	3125	3125	3125	3125	3125	3125
1	3134	3124	3120	3113	3110	3114	3126	3124	3126	3123	3122	3117	3112	3120	3122	3129	**3135**
2	3132	3125	3117	3114	3107	3117	3122	**3135**	3132	3124	3118	3113	3115	3119	3123	3125	3134
3	3126	3117	3121	3107	3109	3117	3119	3124	3131	3126	3121	3108	3110	3121	3128	3128	3128
4	3125	3118	3111	3105	3107	3112	3124	3122	3128	3124	3117	3108	3109	3115	3126	3125	3129
5	3121	3113	3106	3099	3100	3111	3117	3119	3122	3119	3111	3110	3108	3113	3123	3124	3123
6	3121	3107	3103	3096	3098	3101	3111	3119	3116	3115	3108	3106	3100	3103	3109	3121	3120
7	3116	3101	3098	3092	3096	3097	3106	3113	3111	3107	3108	3102	3095	3099	3105	3115	3110
8	3104	3102	3095	3090	3093	3097	3105	3105	3105	3103	3099	3092	3095	3101	3107	3105	3109
9	3099	3097	3086	3086	3081	3087	3093	3099	3106	3102	3099	3091	3090	3093	3101	3104	3103
10	3087	3082	3081	3071	3069	3080	3087	3090	3093	3087	3081	3077	3082	3081	3083	3086	3088
11	3075	3073	3062	3058	3057	3069	3076	3080	3077	3078	3074	3066	3065	3076	3079	3079	3082
12	3060	3052	3054	3049	3048	3052	3057	3059	3068	3065	3056	3054	3053	3060	3063	3063	3062
13	3055	3039	3035	3029	3028	3034	3041	3051	3048	3045	3049	3039	3046	3042	3047	3046	3047
14	3031	3025	3023	3019	3018	3025	3033	3036	3032	3034	3026	3029	3026	3031	3034	3038	3028
15	3025	3010	3003	2998	2999	3003	3015	3018	3026	3018	3017	3012	3014	3016	3019	3015	3019
16	3014	2995	2992	2986	2987	2996	3003	3004	3001	3006	3000	3000	3002	3004	3007	3005	2998
17	2995	2976	2979	2969	2977	2977	2982	2987	2995	2985	2989	2985	2989	2990	2989	2986	2990
18	2983	2965	2963	2960	2957	2966	2974	2976	2974	2972	2974	2976	2970	2970	2979	2975	2972
19	2956	2945	2949	2943	2946	2950	2955	2959	2966	2961	2964	2957	2960	2959	2962	2958	2958
20	2937	2937	2933	2926	2928	2932	2943	2943	2947	2943	2946	2946	2944	2942	2946	2945	2945
25	2823	2819	2814	2807	2814	2818	2818	2827	2823	2827	2824	2831	2832	2832	2830	2825	2816
30	2700	2674	2678	2675	2680	2682	2690	2689	2692	2689	2695	2699	2704	2698	2693	2690	2683
35	2530	2540	2543	2537	2543	2550	2553	2550	2554	2558	2566	2569	2570	2561	2561	2549	2542
40	2352	2353	2348	2353	2354	2360	2358	2359	2362	2372	2379	2388	2388	2392	2376	2367	2351
45	2160	2125	2123	2127	2131	2138	2138	2136	2136	2153	2161	2170	2171	2172	2159	2147	2128
50	1947	1893	1891	1896	1897	1900	1897	1900	1904	1918	1933	1945	1944	1945	1924	1909	1898
55	1707	1648	1646	1651	1657	1655	1656	1653	1661	1678	1696	1708	1710	1708	1687	1668	1660
60	1461	1407	1409	1411	1413	1412	1415	1416	1424	1438	1460	1474	1479	1468	1450	1431	1419
65	1217	1169	1178	1174	1180	1175	1176	1178	1194	1204	1227	1239	1245	1228	1221	1194	1189
70	957	945	948	945	951	945	948	952	968	979	1005	1016	1019	1000	996	968	959
75	692	688	678	680	678	686	682	689	695	719	737	750	753	755	729	715	689
80	456	416	408	408	406	412	410	416	429	447	464	479	481	483	457	444	424
85	229	180	174	171	171	172	175	183	196	209	225	237	241	236	221	207	195
90	41	18	14	10	8	9	13	18	27	34	41	45	46	41	36	30	27
95	3	3	3	2	3	3	3	3	2	3	3	3	2	3	3	3	2
100	2	2	2	2	3	3	2	2	2	2	2	3	2	3	3	2	2
105	3	2	3	3	3	3	2	2	3	2	3	3	3	3	3	3	3
110	3	3	2	3	2	2	3	3	3	3	2	2	3	3	2	3	2
115	2	3	2	3	3	3	3	3	2	3	2	3	3	2	3	3	3
120	2	2	3	3	3	3	3	3	2	3	3	3	2	3	3	2	3
125	3	2	3	3	3	3	3	2	3	2	3	3	3	3	3	3	3
130	3	3	3	4	3	3	3	3	3	3	3	3	3	3	3	3	3
135	3	3	3	3	4	3	4	3	3	4	4	4	4	3	3	4	3
140	3	3	4	3	4	4	4	3	3	4	4	4	4	4	4	4	4
145	4	4	4	4	4	4	4	4	4	4	4	4	4	4	4	4	4
150	4	4	4	4	4	4	4	4	4	4	4	4	4	4	4	4	4
155	4	4	4	4	4	4	4	4	4	4	4	4	4	4	4	4	4
160	4	4	4	4	4	4	4	4	4	4	4	4	4	5	5	4	5
165	5	5	5	4	5	5	5	5	5	5	5	5	5	5	5	5	5
170	6	6	5	5	6	5	6	5	6	6	6	5	6	6	6	6	6
175	6	6	6	6	6	6	6	6	6	7	6	6	6	6	6	7	6
180	6	6	6	6	6	6	6	6	6	6	6	6	6	6	6	6	6

Corrected UGR values generated per CIE 190-2010

UGR Table - Corrected

Reflectances											
Ceiling Cavity		70	70	50	50	30	70	70	50	50	30
Walls		50	30	50	30	30	50	30	50	30	30
Floor Cavity		20	20	20	20	20	20	20	20	20	20
Room Size		UGR Viewed Crosswise					UGR Viewed Endwise				
X=2H	Y=2H	17.3	18.9	17.6	19.2	19.6	17.2	18.9	17.6	19.2	19.5
	3H	19.3	20.8	19.7	21.1	21.5	19.2	20.7	19.6	21.1	21.4
	4H	20.1	21.5	20.5	21.9	22.2	20.0	21.4	20.4	21.8	22.2
	6H	20.7	22.0	21.1	22.4	22.8	20.6	21.9	21.0	22.3	22.6
	8H	20.9	22.2	21.3	22.6	23.0	20.7	22.0	21.2	22.4	22.8
	12H	21.1	22.3	21.5	22.7	23.1	20.8	22.1	21.3	22.5	22.9
4H	2H	18.0	19.4	18.4	19.7	20.1	17.9	19.3	18.3	19.7	20.1
	3H	20.2	21.4	20.6	21.8	22.2	20.2	21.4	20.6	21.8	22.2
	4H	21.2	22.3	21.6	22.7	23.1	21.1	22.2	21.5	22.6	23.0
	6H	21.9	22.9	22.4	23.3	23.8	21.7	22.7	22.2	23.2	23.6
	8H	22.2	23.1	22.6	23.5	24.0	22.0	22.9	22.4	23.3	23.8
	12H	22.4	23.2	22.9	23.7	24.1	22.1	23.0	22.6	23.4	23.9
8H	4H	21.6	22.4	22.0	22.9	23.4	21.4	22.3	21.9	22.8	23.2
	6H	22.4	23.2	22.9	23.7	24.1	22.2	23.0	22.7	23.5	24.0
	8H	22.8	23.4	23.3	23.9	24.4	22.5	23.2	23.0	23.7	24.2
	12H	23.0	23.7	23.6	24.1	24.7	22.8	23.4	23.3	23.9	24.4
12H	4H	21.6	22.4	22.1	22.9	23.3	21.5	22.3	22.0	22.8	23.2
	6H	22.5	23.2	23.0	23.6	24.2	22.3	23.0	22.8	23.5	24.0
	8H	22.9	23.5	23.4	24.0	24.5	22.7	23.3	23.2	23.8	24.3

Maximum UGR = 24.7

THD and PF Test

Model No.	BPANEL-75S5-UNV-D-8SC61-24-G-WH-H	Sample ID.	7712181

Test Method

1. The samples were tested according to the ANSI C82.77-10-2014.
2. The ambient temperature condition was maintained at 25 °C ± 1 °C. The sample measurement was made using a digital power meter and power supply. The sample was operated at rated voltage and stabilized before measurement. The total harmonic distortion were calculated from the digital power meter.

Test Results

CCT Range	Voltage (Vac)	Frequency (Hz)	Current (A)	Power (W)	Power Factor	Current THD	Orientation
0%-3000K	120	60	0.588	70.3	0.9950	8.74%	Horizontal
	277	60	0.256	69.4	0.9792	7.01%	Horizontal
20%-3500K	120	60	0.585	69.9	0.9950	8.73%	Horizontal
	277	60	0.255	69.1	0.9789	7.17%	Horizontal
40%-4000K	120	60	0.585	69.9	0.9950	8.77%	Horizontal
	277	60	0.255	69.0	0.9788	7.14%	Horizontal
60%-5000K	120	60	0.585	69.8	0.9949	8.75%	Horizontal
	277	60	0.255	69.0	0.9788	7.13%	Horizontal
80%-5700K	120	60	0.502	60.0	0.9949	8.80%	Horizontal
	277	60	0.254	69.0	0.9788	7.13%	Horizontal
100%-6500K	120	60	0.587	70.1	0.9950	8.75%	Horizontal
	277	60	0.255	69.3	0.9790	7.12%	Horizontal

In-Situ Temperature Measurement Test

Model No.	BPANEL-75S5-UNV-D-8SC61-24-G-WH-H	Sample ID.	7712181

Test Method

1. In-Situ Temperature Measurement Test is conducted according to the UL 1598-2008, Section 14.
2. The testing was conducted in a room with ambient temperature of 25 °C ± 5 °C. The apparatus construction followed those described in UL1598-2008 for normal temperature testing. Thermocouples were placed on the LED package in the locations indicated by LM-80 report. Thermocouples were placed on the LED driver case in the locations specified by the manufacture if necessary. The temperature was recorded after the lamp was operated by 7.5 hours.
3.The data and photos in LM-80 test report is provided by the customer. The data and photos in driver specification is provided by the customer.

In-Situ Temperature Measurement Test Conditions

Temperature (°C)	Voltage (Vac)	Frequency (Hz)	Current (A)	Power (W)	Power Factor	Orientation
24.5	120	60	0.588	70.3	0.9950	Horizontal

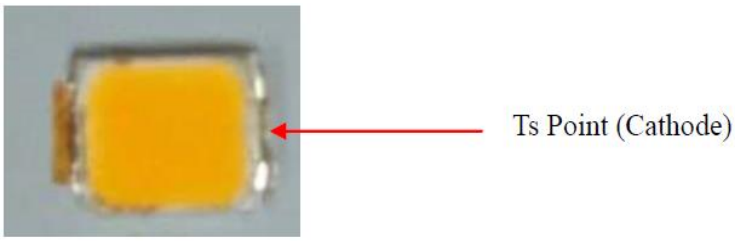
Test Results (LEDs)

Thermocouple Location	Manufacturer Declared Current (mA)	Temperature for Lighting source (°C)		LED Model Number	LM-80 Limit Current (mA)	LM-80 Limit Temp. (°C)
		Test result column 1	Test result (Correct to 25 °C)			
TMP of LEDs	100	35.1	35.6	BXVN-30E-13H-9NV	100	115
Ambient temperature	N/A	24.5	25.0			

Test Results(Driver)

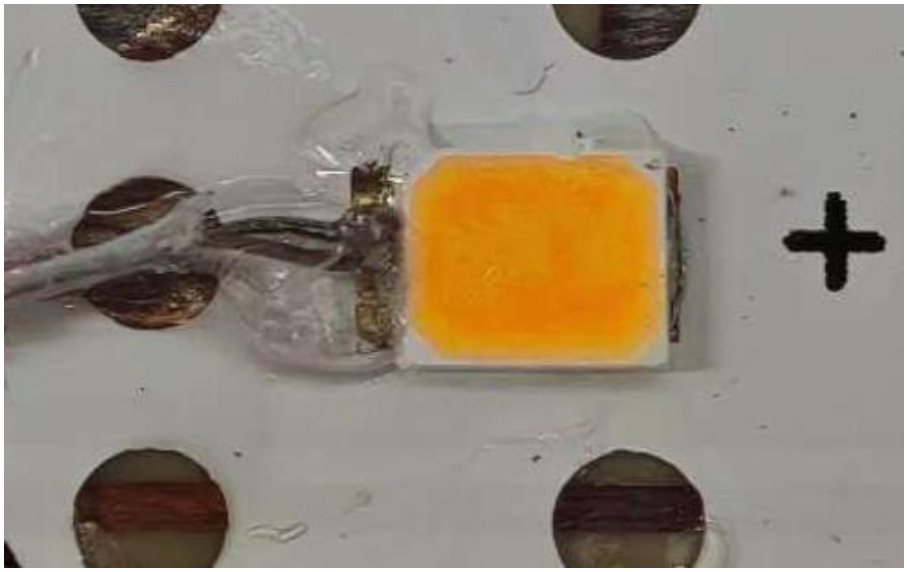
Thermocouple Location	Temperature for Driver (°C)		Driver Model Number	Driver Limit Temp. (°C)
	Test result column 1	Test result (Correct to 25 °C)		
Tc of Driver	56.5	57.0	BQ-65L6-1470-44-M56	90
Ambient temperature	24.5	25.0		

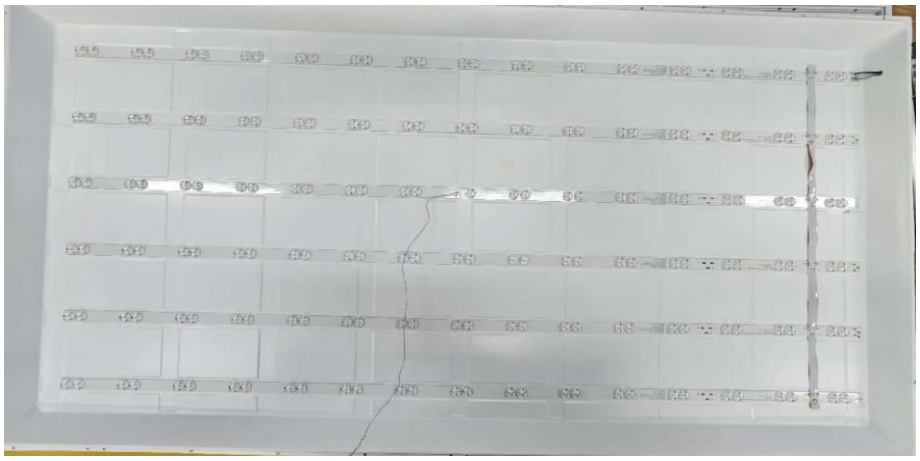
TMP point in LM-80 report



In-Situ Temperature Measurement Test (Cont'd)

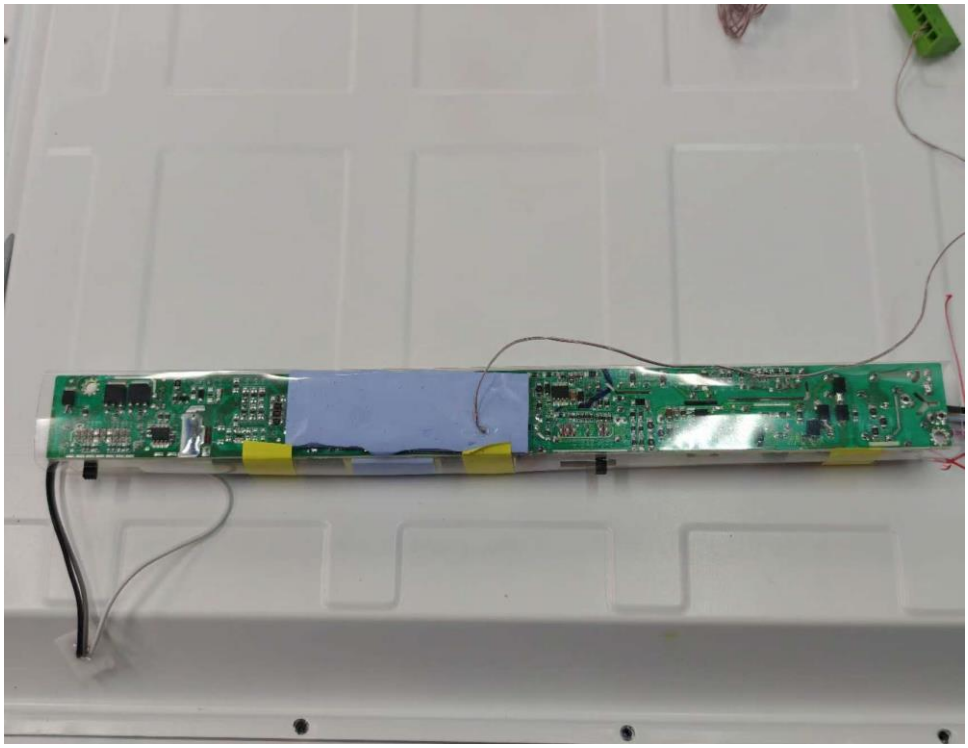
Test Photos for TMP of LED Packages





In-Situ Temperature Measurement Test (Cont'd)

Test Photos for Tc Point of Driver



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